Chapter 3. Integration

The Riemann integral over rectangles and bounded sets, existence via measure zero, Fubini's theorem, rectifiable sets, improper integrals

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Overview

We define the (Riemann) integral of a bounded real-valued function of several variables, first over a rectangle (§10), and characterize integrability: f is integrable iff its set of discontinuities has measure zero (§11). Fubini's theorem reduces integrals to iterated integrals (§12). The integral is then extended to bounded sets (§13), which leads to rectifiable sets and volume (§14), and finally to the extended (improper) integral over open sets (§15).

§10. The Integral over a Rectangle

Motivation. To integrate over regions in $\mathbb{R}^n$ we follow the one-variable Riemann approach: partition a rectangle into subrectangles and compare upper and lower sums. The results here are the basic bookkeeping of refinements.

Definition 10.1 (rectangle, volume)

Let $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ be a rectangle in $\mathbb{R}^n$. The intervals $[a_i, b_i]$ are its **component intervals**; the maximum of $b_i - a_i$ is its **width**; the **volume** of Q is $\mathrm{v}(Q) = (b_1 - a_1)(b_2 - a_2)\cdots(b_n - a_n)$. For $n = 1$ volume and width coincide and are called the **length** of $[a, b]$.

Definition 10.2 (partition)

A **partition** of $[a, b]$ is a finite set P of points of $[a, b]$ containing a and b; we index it as $a = t_0 < t_1 < \cdots < t_k = b$, and each $[t_{i-1}, t_i]$ is a **subinterval determined by** P. A **partition** of a rectangle Q as above is an n-tuple $P = (P_1, \ldots, P_n)$ where P_j is a partition of $[a_j, b_j]$. If I_j is a subinterval determined by P_j, then $R = I_1 \times \cdots \times I_n$ is a **subrectangle determined by** P. The maximum width of these subrectangles is the **mesh** of P.

Definition 10.3 (upper and lower sums)

Let $f : Q \to \mathbb{R}$ be bounded and P a partition of Q. For each subrectangle R determined by P let $m_R(f) = \inf\{f(\mathbf{x}) : \mathbf{x} \in R\}$ and $M_R(f) = \sup\{f(\mathbf{x}) : \mathbf{x} \in R\}$. The **lower sum** and **upper sum** of f determined by P are

$$L(f, P) = \sum_R m_R(f)\ \mathrm{v}(R), \qquad U(f, P) = \sum_R M_R(f)\ \mathrm{v}(R),$$

the sums extending over all subrectangles R determined by P.

If P'' is obtained from $P = (P_1, \ldots, P_n)$ by adjoining points to some or all of the P_j, then P'' is a **refinement** of P. The partition $P'' = (P_1 \cup P_1', \ldots, P_n \cup P_n')$ is a refinement of both P and P', their **common refinement**. Refining increases lower sums and decreases upper sums:

Lemma 10.1

Let P be a partition of the rectangle Q; let $f : Q \to \mathbb{R}$ be a bounded function. If P'' is a refinement of P, then

$$L(f, P) \le L(f, P'') \qquad \text{and} \qquad U(f, P'') \le U(f, P).$$

Proof. It suffices to treat P'' obtained by adjoining a single point q to one component partition, say $P_1 = \{a_1 = t_0 < \cdots < t_k = b_1\}$, with q interior to $[t_{i-1}, t_i]$. Most subrectangles of P are

also subrectangles of P''. The exceptions are those of the form $R_S = [t_{i-1}, t_i] \times S$, where S is a subrectangle of $[a_2, b_2] \times \cdots \times [a_n, b_n]$ determined by $(P_2, \ldots, P_n)$; in the lower sum the term for R_S is replaced by the terms for $R'_S = [t_{i-1}, q] \times S$ and $R''_S = [q, t_i] \times S$. Since $m_{R_S}(f) \le f(\mathbf{x})$ for $\mathbf{x}$ in R'_S and in R''_S, we get $m_{R_S}(f) \le m_{R'_S}(f)$ and $m_{R_S}(f) \le m_{R''_S}(f)$. Because $\mathrm{v}(R_S) = \mathrm{v}(R'_S) + \mathrm{v}(R''_S)$,

$$m_{R_S}(f)\,\mathrm{v}(R_S) \le m_{R'_S}(f)\,\mathrm{v}(R'_S) + m_{R''_S}(f)\,\mathrm{v}(R''_S).$$

Summing over S gives $L(f, P) \le L(f, P'')$. Similarly $U(f, P) \ge U(f, P'')$. ■

Lemma 10.2

Let Q be a rectangle; let $f : Q \to \mathbb{R}$ be a bounded function. If P and P' are any two partitions of Q, then $L(f, P) \le U(f, P')$.

Proof. If $P = P'$, this follows from $m_R(f) \le M_R(f)$. In general let P'' be the common refinement; by Lemma 10.1, $L(f, P) \le L(f, P'') \le U(f, P'') \le U(f, P')$. ■

Definition 10.4 (the integral)

Let $f : Q \to \mathbb{R}$ be bounded. As P ranges over all partitions of Q, define the **lower integral** and **upper integral**

$$\underline{\int_Q} f = \sup_P \{L(f, P)\}, \qquad \overline{\int_Q} f = \inf_P \{U(f, P)\}.$$

They exist since the $L(f, P)$ are bounded above by any $U(f, P')$ and the $U(f, P)$ are bounded below by any $L(f, P')$. If they are equal, f is **integrable** over Q, and the common value is the **integral** $\int_Q f$ or $\int_{\mathbf{x} \in Q} f(\mathbf{x})$.

Example 10.1

If $f : [a, b] \to \mathbb{R}$ is non-negative and bounded, $L(f, P)$ is the total area of rectangles inscribed in the region between the graph of f and the x-axis, and $U(f, P)$ the total area of rectangles circumscribed about it. The lower integral is the "inner area" and the upper integral the "outer area" of the region; f is integrable when they agree. Similarly for $Q \subset \mathbb{R}^2$, with boxes in place of rectangles.

Example 10.2

Let $f : [0, 1] \to \mathbb{R}$, $f(x) = 0$ if x is rational and $f(x) = 1$ if x is irrational. Every subinterval R contains rational and irrational numbers, so $m_R(f) = 0$, $M_R(f) = 1$, $L(f, P) = 0$ and $U(f, P) = \sum_R \mathrm{v}(R) = 1$ for every P. The lower integral is 0 and the upper integral 1: f is not integrable.

Theorem 10.3 (The Riemann condition)

Let Q be a rectangle; let $f : Q \to \mathbb{R}$ be a bounded function. Then

$$\underline{\int_Q} f \le \overline{\int_Q} f;$$

equality holds if and only if given $\epsilon > 0$ there exists a corresponding partition P of Q for which $U(f, P) - L(f, P) < \epsilon$.

Proof. Fix P'. Since $L(f,P) \le U(f,P')$ for all P, the lower integral is $\le U(f,P')$; as P' is arbitrary, the lower integral is $\le$ the upper integral.

Suppose they are equal. Choose P with $L(f,P)$ within $\epsilon/2$ of $\int_Q f$, and P' with $U(f,P')$ within $\epsilon/2$ of $\int_Q f$; let P'' be their common refinement. Since $L(f,P) \le L(f,P'') \le \int_Q f \le U(f,P'') \le U(f,P')$, the sums for P'' are within ϵ of each other. Conversely, if the upper and lower integrals differ, let ϵ be their difference; then for every P, $L(f,P) \le \underline{\int}_Q f < \overline{\int}_Q f \le U(f,P)$, so $U(f,P) - L(f,P) \ge \epsilon$. ■

Theorem 10.4

Every constant function $f(\mathbf{x}) = c$ is integrable. Indeed, if Q is a rectangle and if P is a partition of Q, then

$$\int_Q c = c \cdot \mathrm{v}(Q) = c \sum_R \mathrm{v}(R),$$

where the summation extends over all subrectangles determined by P.

Proof. $m_R(f) = c = M_R(f)$ for every R, so $L(f,P) = c\sum_R \mathrm{v}(R) = U(f,P)$ and the Riemann condition holds trivially; $\int_Q c$ lies between them, so equals $c\sum_R \mathrm{v}(R)$. This holds for every P; for the trivial partition (whose only subrectangle is Q) it gives $c \cdot \mathrm{v}(Q)$. ■

Corollary 10.5

Let Q be a rectangle in $\mathbb{R}^n$; let $\{Q_1, \ldots, Q_k\}$ be a finite collection of rectangles that covers Q. Then $\mathrm{v}(Q) \le \sum_{i=1}^k \mathrm{v}(Q_i)$.

Proof. Choose a rectangle Q' containing $Q, Q_1, \ldots, Q_k$, and use the end points of the component intervals of all these rectangles to define a partition P of Q'. Each of $Q, Q_1, \ldots, Q_k$ is a union of subrectangles determined by P. By Theorem 10.4 (applied to the partition of Q induced by P), $\mathrm{v}(Q) = \sum_{R \subset Q} \mathrm{v}(R)$. Each such R lies in at least one Q_i, so

$$\sum_{R \subset Q} \mathrm{v}(R) \le \sum_{i=1}^k \sum_{R \subset Q_i} \mathrm{v}(R) = \sum_{i=1}^k \mathrm{v}(Q_i).$$

■

Notation

For $n = 1$ we often write $\int_a^b f$ or $\int_{x=a}^{x=b} f(x)$ for $\int_{[a,b]} f$. The calculus notation $\int_a^b f(x)\,dx$, in which "dx" has no independent meaning, is avoided for now; later dx will be given a meaning. This definition is due to Darboux; an equivalent one due to Riemann uses sums $\sum_R f(\mathbf{x}_R)\,\mathrm{v}(R)$. Either way, it is called the **Riemann integral**.

§11. Existence of the Integral

Motivation. Which bounded functions are integrable? The answer is clean: exactly those whose discontinuities form a set of measure zero. This criterion is what lets us integrate over sets with complicated boundaries later.

Definition 11.1 (measure zero)

A subset A of $\mathbb{R}^n$ has **measure zero** in $\mathbb{R}^n$ if for every $\epsilon > 0$ there is a covering $Q_1, Q_2, \ldots$ of A by countably many rectangles such that $\sum_{i=1}^{\infty} \mathrm{v}(Q_i) < \epsilon$ ("the total volume of the Q_i is less than ϵ").

Theorem 11.1

(a) If $B \subset A$ and A has measure zero in $\mathbb{R}^n$, then so does B.

(b) Let A be the union of the countable collection of sets $A_1, A_2, \ldots$. If each A_i has measure zero in $\mathbb{R}^n$, so does A.

(c) A set A has measure zero in $\mathbb{R}^n$ if and only if for every $\epsilon > 0$ there is a countable covering of A by open rectangles $\operatorname{Int} Q_1, \operatorname{Int} Q_2, \ldots$ such that $\sum_{i=1}^{\infty} \mathrm{v}(Q_i) < \epsilon$.

(d) If Q is a rectangle in $\mathbb{R}^n$, then $\operatorname{Bd} Q$ has measure zero in $\mathbb{R}^n$ but Q does not.

Proof. (a) is immediate.

(b) Cover A_j by countably many rectangles $Q_{1j}, Q_{2j}, \ldots$ of total volume less than $\epsilon/2^j$. The collection $\{Q_{ij}\}$ is countable, covers A, and has total volume less than $\sum_j \epsilon/2^j = \epsilon$.

(c) If open rectangles $\operatorname{Int} Q_i$ cover A, so do the Q_i; this gives "if". Conversely, cover A by rectangles $Q_1', Q_2', \ldots$ of total volume less than $\epsilon/2$, and for each i choose a rectangle Q_i with $Q_i' \subset \operatorname{Int} Q_i$ and $\mathrm{v}(Q_i) \leq 2\,\mathrm{v}(Q_i')$ (possible since v is a continuous function of the end points of the component intervals). Then the $\operatorname{Int} Q_i$ cover A and $\sum \mathrm{v}(Q_i) < \epsilon$.

(d) Let $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$. The subset of Q where $x_i = a_i$ is one of the ith **faces** of Q, the other where $x_i = b_i$. Each face has measure zero: e.g. the face $x_i = a_i$ is covered by the single rectangle $[a_1, b_1] \times \cdots \times [a_i, a_i + \delta] \times \cdots \times [a_n, b_n]$, whose volume is as small as desired. $\operatorname{Bd} Q$ is the union of the finitely many faces, so it has measure zero. Suppose Q had measure zero, and let $\epsilon = \mathrm{v}(Q)$. Cover Q by open rectangles $\operatorname{Int} Q_1, \operatorname{Int} Q_2, \ldots$ with $\sum \mathrm{v}(Q_i) < \epsilon$. Since Q is compact, finitely many cover Q, say $\operatorname{Int} Q_1, \ldots, \operatorname{Int} Q_k$; but $\sum_{i=1}^{k} \mathrm{v}(Q_i) < \epsilon = \mathrm{v}(Q)$ contradicts Corollary 10.5. ■

Example 11.1

Allowing *countably infinite* coverings is essential. The set A of rationals in $I = [0, 1]$ is a countable union of one-point sets, hence has measure zero by (b). But A cannot be covered by finitely many intervals of total length less than $\epsilon < 1$: if $I_1, \ldots, I_k$ cover A, their union B is closed (a finite union of closed sets) and contains all rationals of I, hence all their limit points, i.e. all of I; by Corollary 10.5, $\sum \mathrm{v}(I_i) \geq \mathrm{v}(I) = 1$.

Theorem 11.2

Let Q be a rectangle in $\mathbb{R}^n$; let $f : Q \to \mathbb{R}$ be a bounded function. Let D be the set of points of Q at which f fails to be continuous. Then $\int_Q f$ exists if and only if D has measure zero in $\mathbb{R}^n$.

Proof. Choose M with $|f(\mathbf{x})| \leq M$ for $\mathbf{x} \in Q$.

Step 1. ("If") Assume D has measure zero. Given $\epsilon > 0$, let $\epsilon' = \epsilon/(2M + 2\,\mathrm{v}(Q))$. Cover D by open rectangles $\operatorname{Int} Q_1, \operatorname{Int} Q_2, \ldots$ with total volume less than ϵ' (Theorem 11.1(c)). For each $\mathbf{a} \in Q - D$ choose an open rectangle $\operatorname{Int} Q_{\mathbf{a}}$ containing $\mathbf{a}$ such that $|f(\mathbf{x}) - f(\mathbf{a})| < \epsilon'$ for $\mathbf{x} \in Q_{\mathbf{a}} \cap Q$ (continuity at $\mathbf{a}$). The open sets $\operatorname{Int} Q_i$ and $\operatorname{Int} Q_{\mathbf{a}}$ cover the compact set Q, so finitely many do:

$\operatorname{Int} Q_1, \dots, \operatorname{Int} Q_k, \operatorname{Int} Q_{\mathbf{a}_1}, \dots, \operatorname{Int} Q_{\mathbf{a}_l}$. Write $Q'_j = Q_{\mathbf{a}_j}$. Then

$$\sum_{i=1}^{k} \mathrm{v}(Q_i) < \epsilon', \tag{1}$$

$$|f(\mathbf{x}) - f(\mathbf{y})| \le 2\epsilon' \quad \text{for } \mathbf{x}, \mathbf{y} \in Q'_j \cap Q. \tag{2}$$

Replace each Q_i and Q'_j by its intersection with Q; (1), (2) persist. Use the end points of the component intervals of $Q_1, \dots, Q_k, Q'_1, \dots, Q'_l$ to define a partition P of Q; each Q_i, Q'_j is a union of subrectangles determined by P. Divide these subrectangles into two disjoint collections $\mathcal{R}$ and $\mathcal{R}'$, with each $R \in \mathcal{R}$ lying in some Q_i and each $R \in \mathcal{R}'$ lying in some Q'_j. Since $|f(\mathbf{x}) - f(\mathbf{y})| \le 2M$ in general and $\le 2\epsilon'$ on $R \in \mathcal{R}'$,

$$\sum_{R\in\mathcal{R}} (M_R(f) - m_R(f))\,\mathrm{v}(R) \le 2M \sum_{R\in\mathcal{R}} \mathrm{v}(R), \qquad \sum_{R\in\mathcal{R}'} (M_R(f) - m_R(f))\,\mathrm{v}(R) \le 2\epsilon' \sum_{R\in\mathcal{R}'} \mathrm{v}(R).$$

Now $\sum_{R\in\mathcal{R}} \mathrm{v}(R) \le \sum_{i=1}^{k} \sum_{R\subset Q_i} \mathrm{v}(R) = \sum_i \mathrm{v}(Q_i) < \epsilon'$ and $\sum_{R\in\mathcal{R}'} \mathrm{v}(R) \le \sum_{R\subset Q} \mathrm{v}(R) = \mathrm{v}(Q)$. Thus $U(f,P) - L(f,P) < 2M\epsilon' + 2\epsilon'\,\mathrm{v}(Q) = \epsilon$, and f is integrable by the Riemann condition.

Step 2. Oscillation. For $\mathbf{a} \in Q$ and $\delta > 0$ let $A_\delta = \{f(\mathbf{x}) : \mathbf{x} \in Q,\ |\mathbf{x} - \mathbf{a}| < \delta\}$, $M_\delta(f) = \sup A_\delta$, $m_\delta(f) = \inf A_\delta$. The **oscillation** of f at $\mathbf{a}$ is

$$\nu(f;\mathbf{a}) = \inf_{\delta>0} [M_\delta(f) - m_\delta(f)] \ \ge 0.$$

f is continuous at $\mathbf{a}$ iff $\nu(f;\mathbf{a}) = 0$. If f is continuous at $\mathbf{a}$, given ϵ choose δ with $|f(\mathbf{x}) - f(\mathbf{a})| < \epsilon$ for $|\mathbf{x} - \mathbf{a}| < \delta$; then $M_\delta(f) \le f(\mathbf{a}) + \epsilon$, $m_\delta(f) \ge f(\mathbf{a}) - \epsilon$, so $\nu(f;\mathbf{a}) \le 2\epsilon$. Conversely if $\nu(f;\mathbf{a}) = 0$, given ϵ there is δ with $M_\delta(f) - m_\delta(f) < \epsilon$; for $|\mathbf{x} - \mathbf{a}| < \delta$ both $f(\mathbf{x})$ and $f(\mathbf{a})$ lie in $[m_\delta(f), M_\delta(f)]$, so $|f(\mathbf{x}) - f(\mathbf{a})| < \epsilon$.

Step 3. ("Only if") Assume f is integrable. Let $D_m = \{\mathbf{a} : \nu(f;\mathbf{a}) \ge 1/m\}$; by Step 2, $D = \bigcup_m D_m$, so it suffices to show each D_m has measure zero. Fix m and $\epsilon > 0$. Choose P with $U(f,P) - L(f,P) < \epsilon/2m$. Let D'_m be the points of D_m lying on $\operatorname{Bd} R$ for some subrectangle R determined by P, and D''_m the rest of D_m. D'_m lies in the finite union $\bigcup_R \operatorname{Bd} R$, which has measure zero, so it can be covered by countably many rectangles of total volume $< \epsilon/2$. Let $R_1, \dots, R_k$ be the subrectangles that contain points of D''_m. If $\mathbf{a} \in D''_m \cap R_i$, then $\mathbf{a} \notin \operatorname{Bd} R_i$, so some cubical neighborhood of radius δ about $\mathbf{a}$ lies in R_i, and

$$1/m \le \nu(f;\mathbf{a}) \le M_\delta(f) - m_\delta(f) \le M_{R_i}(f) - m_{R_i}(f).$$

Multiplying by $\mathrm{v}(R_i)$ and summing: $\sum_{i=1}^{k} (1/m)\,\mathrm{v}(R_i) \le U(f,P) - L(f,P) < \epsilon/2m$, so $R_1, \dots, R_k$ cover D''_m with total volume $< \epsilon/2$. ■

Theorem 11.3

Let Q be a rectangle in $\mathbb{R}^n$; let $f : Q \to \mathbb{R}$; assume f is integrable over Q.

(a) If f vanishes except on a set of measure zero, then $\int_Q f = 0$.

(b) If f is non-negative and if $\int_Q f = 0$, then f vanishes except on a set of measure zero.

Proof. (a) Suppose f vanishes except on a set E of measure zero. A subrectangle R determined by a partition P is not contained in E (Theorem 11.1(d),(a)), so f vanishes at some point of R, whence $m_R(f) \le 0 \le M_R(f)$. Hence $L(f,P) \le 0 \le U(f,P)$ for all P, so $\underline{\int}_Q f \le 0 \le \overline{\int}_Q f$; since $\int_Q f$ exists, it is 0.

(b) Suppose $f \ge 0$ and $\int_Q f = 0$. We show: if f is continuous at $\mathbf{a}$, then $f(\mathbf{a}) = 0$; then f vanishes except on its set of discontinuities, which has measure zero by Theorem 11.2. Suppose $f(\mathbf{a}) > 0$

and let $\epsilon = f(\mathbf{a})$; choose δ with $f(\mathbf{x}) > \epsilon/2$ for $|\mathbf{x} - \mathbf{a}| < \delta$, $\mathbf{x} \in Q$. Take a partition P of mesh less than δ; if R_0 is a subrectangle containing $\mathbf{a}$, then $m_{R_0}(f) \geq \epsilon/2$, and $m_R(f) \geq 0$ for all R. So $L(f, P) \geq (\epsilon/2)\,\mathrm{v}(R_0) > 0$, contradicting $L(f, P) \leq \int_Q f = 0$. ■

Example 11.2

The assumption that $\int_Q f$ exists is necessary in (a): the function f on $I = [0, 1]$ with $f(x) = 1$ for x rational and 0 otherwise vanishes except on a set of measure zero, but $\int_I f$ does not exist (cf. Example 10.2).

§12. Evaluation of the Integral

Motivation. Integrals over rectangles are computed by integrating one variable at a time. Fubini's theorem justifies this and reduces multiple integrals to single ones.

Theorem 12.1 (Fundamental theorem of calculus)

(a) If f is continuous on $[a, b]$, and if $F(x) = \int_a^x f$ for $x \in [a, b]$, then $F'(x)$ exists and equals $f(x)$.
(b) If f is continuous on $[a, b]$, and if g is a function such that $g'(x) = f(x)$ for $x \in [a, b]$, then $\int_a^b f = g(b) - g(a)$.

(At the end points, F' and g' mean one-sided derivatives.) In short: $D \int_a^x f = f(x)$ and $\int_a^x Dg = g(x) - g(a)$.

Proof [Supplement]. (a) By additivity (Theorem 13.3(d), or directly for intervals), $F(x + h) - F(x) = \int_x^{x+h} f$ for $h > 0$ (similarly for $h < 0$). By comparison, this lies between $h \min f$ and $h \max f$ over $[x, x + h]$, so $\frac{F(x+h)-F(x)}{h}$ lies between $f(s)$ and $f(t)$ for some $s, t \in [x, x + h]$; by continuity it tends to $f(x)$ as $h \to 0$.

(b) $F - g$ has derivative 0 on $[a, b]$, so it is constant by the mean-value theorem (Theorem 6.1); hence $\int_a^b f = F(b) - F(a) = g(b) - g(a)$. ■

So we can compute the integral of a continuous f by finding an **antiderivative** g, $g' = f$; part (a) shows one always exists in theory.

For n-dimensional integrals we reduce to a sequence of lower-dimensional integrals. For continuous f on $Q = [a, b] \times [c, d]$ calculus uses

$$\int_Q f = \int_{x=a}^{x=b} \int_{y=c}^{y=d} f(x, y)$$

without proof. When f is merely integrable, the inner integral may fail to exist for some x (the function can behave badly along a single vertical line without affecting $\int_Q f$). We therefore replace the inner integral by the corresponding lower (or upper) integral, which always exists; this gives a correct general theorem.

Theorem 12.2 (Fubini's theorem)

Let $Q = A \times B$, where A is a rectangle in $\mathbb{R}^k$ and B is a rectangle in $\mathbb{R}^n$. Let $f : Q \to \mathbb{R}$ be a bounded function; write f in the form $f(\mathbf{x}, \mathbf{y})$ for $\mathbf{x} \in A$ and $\mathbf{y} \in B$. For each $\mathbf{x} \in A$, consider

the lower and upper integrals

$$\underline{\int_{\mathbf{y}\in B}} f(\mathbf{x},\mathbf{y}) \qquad \text{and} \qquad \overline{\int_{\mathbf{y}\in B}} f(\mathbf{x},\mathbf{y}).$$

If f is integrable over Q, then these two functions of $\mathbf{x}$ are integrable over A, and

$$\int_Q f = \int_{\mathbf{x}\in A} \underline{\int_{\mathbf{y}\in B}} f(\mathbf{x},\mathbf{y}) = \int_{\mathbf{x}\in A} \overline{\int_{\mathbf{y}\in B}} f(\mathbf{x},\mathbf{y}).$$

Proof. Let $\underline{I}(\mathbf{x}) = \underline{\int}_{\mathbf{y}\in B} f(\mathbf{x},\mathbf{y})$ and $\overline{I}(\mathbf{x}) = \overline{\int}_{\mathbf{y}\in B} f(\mathbf{x},\mathbf{y})$. A partition P of Q consists of a partition P_A of A and P_B of B; if R_A, R_B are general subrectangles determined by P_A, P_B, then $R_A \times R_B$ is the general subrectangle determined by P.

Step 1. $L(f,P) \le L(\underline{I}, P_A)$. Fix $R_A \times R_B$ and $\mathbf{x}_0 \in R_A$. Then $m_{R_A\times R_B}(f) \le f(\mathbf{x}_0,\mathbf{y})$ for all $\mathbf{y} \in R_B$, so $m_{R_A\times R_B}(f) \le m_{R_B}(f(\mathbf{x}_0,\mathbf{y}))$. Multiply by $\mathrm{v}(R_B)$ and sum over R_B:

$$\sum_{R_B} m_{R_A\times R_B}(f)\,\mathrm{v}(R_B) \le L(f(\mathbf{x}_0,\cdot),P_B) \le \underline{I}(\mathbf{x}_0).$$

This holds for each $\mathbf{x}_0 \in R_A$, so $\sum_{R_B} m_{R_A\times R_B}(f)\,\mathrm{v}(R_B) \le m_{R_A}(\underline{I})$. Multiply by $\mathrm{v}(R_A)$ and sum over R_A; since $\mathrm{v}(R_A)\,\mathrm{v}(R_B) = \mathrm{v}(R_A \times R_B)$, $L(f,P) \le L(\underline{I}, P_A)$.

Step 2. $U(f,P) \ge U(\overline{I}, P_A)$, by the symmetric argument: for $\mathbf{x}_0 \in R_A$, $M_{R_A\times R_B}(f) \ge M_{R_B}(f(\mathbf{x}_0,\mathbf{y}))$, so $\sum_{R_B} M_{R_A\times R_B}(f)\,\mathrm{v}(R_B) \ge U(f(\mathbf{x}_0,\cdot),P_B) \ge \overline{I}(\mathbf{x}_0)$; hence the left side is $\ge M_{R_A}(\overline{I})$, and summing gives the claim.

Step 3. Hence the diagram

$$L(f,P) \le L(\underline{I},P_A) \le \begin{Bmatrix} U(\underline{I},P_A) \\ L(\overline{I},P_A) \end{Bmatrix} \le U(\overline{I},P_A) \le U(f,P),$$

in which the two middle terms are each between $L(\underline{I},P_A)$ and $U(\overline{I},P_A)$: the first and last inequalities are Steps 1 and 2, and the middle ones follow from $L(h,P_A) \le U(h,P_A)$ and $\underline{I}(\mathbf{x}) \le \overline{I}(\mathbf{x})$.

Step 4. Since f is integrable, given $\epsilon > 0$ choose P with $U(f,P) - L(f,P) < \epsilon$. Then the upper and lower sums for $\underline{I}$ are within ϵ of each other, and likewise for $\overline{I}$; so both are integrable over A. The numbers $\int_A \underline{I}$, $\int_A \overline{I}$, $\int_Q f$ all lie between the extreme ends of the diagram. As ϵ is arbitrary, they are equal. ∎

One may equally compute $\int_Q f$ by first taking the lower (or upper) integral with respect to $\mathbf{x}$ and then integrating in $\mathbf{y}$.

Corollary 12.3

Let $Q = A \times B$, where A is a rectangle in $\mathbb{R}^k$ and B is a rectangle in $\mathbb{R}^n$. Let $f : Q \to \mathbb{R}$ be a bounded function. If $\int_Q f$ exists, and if $\int_{\mathbf{y}\in B} f(\mathbf{x},\mathbf{y})$ exists for each $\mathbf{x} \in A$, then

$$\int_Q f = \int_{\mathbf{x}\in A} \int_{\mathbf{y}\in B} f(\mathbf{x},\mathbf{y}).$$

Proof [Supplement]. If the inner integral exists, it equals both $\underline{I}(\mathbf{x})$ and $\overline{I}(\mathbf{x})$; apply Fubini's theorem. ∎

Corollary 12.4

Let $Q = I_1 \times \cdots \times I_n$, where I_j is a closed interval in $\mathbb{R}$ for each j. If $f : Q \to \mathbb{R}$ is continuous, then

$$\int_Q f = \int_{x_1 \in I_1} \cdots \int_{x_n \in I_n} f(x_1, \ldots, x_n).$$

Proof [Supplement]. Continuous functions on rectangles are integrable (Theorem 11.2). Write $Q = I_1 \times Q'$ with $Q' = I_2 \times \cdots \times I_n$. For fixed x_1, $f(x_1, \cdot)$ is continuous on Q', hence integrable, so Corollary 12.3 gives $\int_Q f = \int_{x_1 \in I_1} \int_{Q'} f(x_1, \cdot)$. Apply induction on n to the inner integral. ■

§13. The Integral over a Bounded Set

Motivation. Regions such as discs are not rectangles. We extend f by zero outside the set and integrate over a rectangle; integrability then depends on the boundary of the set, which must have measure zero.

To integrate over non-rectangular regions (mass of a circular plate, center of gravity of a spherical cap, ...) we generalize the definition.

Definition 13.1 (integral over a bounded set)

Let S be a bounded set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$ a bounded function. Define $f_S : \mathbb{R}^n \to \mathbb{R}$ by $f_S(\mathbf{x}) = f(\mathbf{x})$ for $\mathbf{x} \in S$ and $f_S(\mathbf{x}) = 0$ otherwise. Choose a rectangle Q containing S and define

$$\int_S f = \int_Q f_S,$$

provided the latter integral exists.

Lemma 13.1

Let Q and Q' be two rectangles in $\mathbb{R}^n$. If $f : \mathbb{R}^n \to \mathbb{R}$ is a bounded function that vanishes outside $Q \cap Q'$, then

$$\int_Q f = \int_{Q'} f;$$

one integral exists if and only if the other does.

Proof. First suppose $Q \subset Q'$. Let E be the set of points of $\operatorname{Int} Q$ where f is discontinuous. Both $f : Q \to \mathbb{R}$ and $f : Q' \to \mathbb{R}$ are continuous except at points of E and possibly points of $\operatorname{Bd} Q$, which has measure zero; so (Theorem 11.2) existence of either integral is equivalent to E having measure zero.

Suppose both exist. Let P be a partition of Q' and P'' its refinement obtained by adjoining the end points of the component intervals of Q. Then Q is a union of subrectangles R determined by P''. If R is not contained in Q, f vanishes at some point of R, so $m_R(f) \le 0$. Hence

$$L(f, P) \le L(f, P'') \le \sum_{R \subset Q} m_R(f)\, \mathrm{v}(R) \le \int_Q f.$$

Similarly $U(f, P) \ge \int_Q f$. As P is an arbitrary partition of Q', $\int_{Q'} f = \int_Q f$. For arbitrary Q, Q' choose Q'' containing both; then $\int_Q f = \int_{Q''} f = \int_{Q'} f$. ■

Lemma 13.2

Let S be a subset of $\mathbb{R}^n$; let $f, g : S \to \mathbb{R}$. Let $F, G : S \to \mathbb{R}$ be defined by

$$F(\mathbf{x}) = \max\{f(\mathbf{x}), g(\mathbf{x})\}, \qquad G(\mathbf{x}) = \min\{f(\mathbf{x}), g(\mathbf{x})\}.$$

(a) If f and g are continuous at $\mathbf{x}_0$, so are F and G.

(b) If f and g are integrable over S, so are F and G.

Proof. (a) If $f(\mathbf{x}_0) = g(\mathbf{x}_0) = r$, then given ϵ choose δ with $|f(\mathbf{x}) - r| < \epsilon$ and $|g(\mathbf{x}) - r| < \epsilon$ for $|\mathbf{x} - \mathbf{x}_0| < \delta$, $\mathbf{x} \in S$; then automatically $|F(\mathbf{x}) - F(\mathbf{x}_0)| < \epsilon$ and $|G(\mathbf{x}) - G(\mathbf{x}_0)| < \epsilon$. If $f(\mathbf{x}_0) > g(\mathbf{x}_0)$, then by continuity $f(\mathbf{x}) - g(\mathbf{x}) > 0$ on $U \cap S$ for a neighborhood U of $\mathbf{x}_0$, where $F = f$ and $G = g$; so F, G are continuous at $\mathbf{x}_0$. Similarly if $f(\mathbf{x}_0) < g(\mathbf{x}_0)$.

(b) Let $Q \supset S$. f_S, g_S are continuous on Q except on sets D, E of measure zero. Since $F_S = \max\{f_S, g_S\}$ and $G_S = \min\{f_S, g_S\}$, by (a) they are continuous except on $D \cup E$, which has measure zero; they are bounded, hence integrable over Q. ■

Theorem 13.3 (Properties of the integral)

Let S be a bounded set in $\mathbb{R}^n$; let $f, g : S \to \mathbb{R}$ be bounded functions.

(a) (Linearity) If f and g are integrable over S, so is $af + bg$, and $\int_S (af + bg) = a \int_S f + b \int_S g$.

(b) (Comparison) Suppose f and g are integrable over S. If $f(\mathbf{x}) \le g(\mathbf{x})$ for $\mathbf{x} \in S$, then $\int_S f \le \int_S g$. Furthermore, $|f|$ is integrable over S and $|\int_S f| \le \int_S |f|$.

(c) (Monotonicity) Let $T \subset S$. If f is non-negative on S and integrable over T and S, then $\int_T f \le \int_S f$.

(d) (Additivity) If $S = S_1 \cup S_2$ and f is integrable over S_1 and S_2, then f is integrable over S and $S_1 \cap S_2$; furthermore

$$\int_S f = \int_{S_1} f + \int_{S_2} f - \int_{S_1 \cap S_2} f.$$

Proof. (a) Since $(af + bg)_S = af_S + bg_S$, it suffices to treat integrals over a rectangle Q. If f, g are integrable over Q, they are continuous except on sets D, E of measure zero; $af + bg$ is continuous except on $D \cup E$, so it is integrable. Suppose first $a, b \ge 0$. For any partition P'' and subrectangle R, $am_R(f) + bm_R(g) \le af(\mathbf{x}) + bg(\mathbf{x})$ for $\mathbf{x} \in R$, so $am_R(f) + bm_R(g) \le m_R(af + bg)$, hence

$$aL(f, P'') + bL(g, P'') \le L(af + bg, P'') \le \int_Q (af + bg),$$

and similarly $aU(f, P'') + bU(g, P'') \ge \int_Q (af + bg)$. For arbitrary P, P' with common refinement P'', $aL(f, P) + bL(g, P') \le \int_Q (af + bg) \le aU(f, P) + bU(g, P')$. The number $a \int_Q f + b \int_Q g$ also lies between these ends for all P, P', so it equals $\int_Q (af + bg)$. It remains to show $\int_Q (-f) = -\int_Q f$: for $\mathbf{x} \in R$, $-M_R(f) \le -f(\mathbf{x}) \le -m_R(f)$, so $-M_R(f) \le m_R(-f)$ and $M_R(-f) \le -m_R(f)$; hence $-U(f, P) \le L(-f, P) \le \int_Q (-f) \le U(-f, P) \le -L(f, P)$, and $-\int_Q f$ also lies between the ends.

(b) It suffices to treat a rectangle Q. If $f \le g$ on Q, then $m_R(f) \le m_R(g)$, so $L(f, P) \le L(g, P) \le \int_Q g$ for every P, whence $\int_Q f \le \int_Q g$. $|f| = \max\{f, -f\}$ is integrable by Lemma 13.2; apply comparison to $-|f| \le f \le |f|$.

(c) $f_T(\mathbf{x}) \le f_S(\mathbf{x})$ for all $\mathbf{x}$; apply comparison.

(d) Let $T = S_1 \cap S_2$. First suppose $f \ge 0$ on S. With $Q \supset S$, f_{S_1}, f_{S_2} are integrable over Q, and $f_S = \max\{f_{S_1}, f_{S_2}\}$, $f_T = \min\{f_{S_1}, f_{S_2}\}$ are integrable by Lemma 13.2. In general put $f_+ = \max\{f, 0\}$, $f_- = \max\{-f, 0\}$; these are integrable over S_1 and S_2 (Lemma 13.2), hence over

S and T by the special case, and so is $f = f_+ - f_-$ by linearity. The formula follows by linearity from $f_S(\mathbf{x}) = f_{S_1}(\mathbf{x}) + f_{S_2}(\mathbf{x}) - f_T(\mathbf{x})$. ■

Corollary 13.4

Let $S_1, \ldots, S_k$ be bounded sets in $\mathbb{R}^n$; assume $S_i \cap S_j$ has measure zero whenever $i \neq j$. Let $S = S_1 \cup \cdots \cup S_k$. If $f : S \to \mathbb{R}$ is integrable over each set S_i, then f is integrable over S and

$$\int_S f = \int_{S_1} f + \cdots + \int_{S_k} f.$$

Proof. For $k = 2$ this is additivity, since $\int_{S_1 \cap S_2} f$ exists and vanishes by Theorem 11.3. Induction gives the general case (note $(S_1 \cup \cdots \cup S_{k-1}) \cap S_k = \bigcup_{i<k}(S_i \cap S_k)$ has measure zero). ■

So far f was only required to be bounded; continuity could not be required, since f_S need not be continuous at points of $\operatorname{Bd} S$ even when f is continuous on S. Our main interest, however, is in continuous integrands.

Convention

Henceforth, we restrict ourselves in studying integration theory to the integration of continuous functions $f : S \to \mathbb{R}$.

Theorem 13.5

Let S be a bounded set in $\mathbb{R}^n$; let $f : S \to \mathbb{R}$ be a bounded continuous function. Let E be the set of points $\mathbf{x}_0$ of $\operatorname{Bd} S$ for which the condition

$$\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = 0$$

fails to hold. If E has measure zero, then f is integrable over S.

Proof. We show f_S is continuous at every $\mathbf{x}_0 \notin E$; then Theorem 11.2 applies. If $\mathbf{x}_0 \in \operatorname{Int} S$, f and f_S agree near $\mathbf{x}_0$, and f is continuous there. If $\mathbf{x}_0 \in \operatorname{Ext} S$, f_S vanishes near $\mathbf{x}_0$. Suppose $\mathbf{x}_0 \in \operatorname{Bd} S - E$ ($\mathbf{x}_0$ may or may not be in S). Then $f(\mathbf{x}) \to 0$ as $\mathbf{x} \to \mathbf{x}_0$ through S; since f is continuous, $f(\mathbf{x}_0) = 0$ if $\mathbf{x}_0 \in S$. As $f_S(\mathbf{x})$ is either $f(\mathbf{x})$ or 0, $f_S(\mathbf{x}) \to 0$ as $\mathbf{x} \to \mathbf{x}_0$. And $f_S(\mathbf{x}_0) = 0$: if $\mathbf{x}_0 \notin S$ by definition, if $\mathbf{x}_0 \in S$ because $f(\mathbf{x}_0) = 0$. ■

The converse also holds (we shall not need it).

Theorem 13.6

Let S be a bounded set in $\mathbb{R}^n$; let $f : S \to \mathbb{R}$ be a bounded continuous function; let $A = \operatorname{Int} S$. If f is integrable over S, then f is integrable over A, and $\int_S f = \int_A f$.

Proof.

Step 1. If f_S is continuous at $\mathbf{x}_0$, then f_A is continuous at $\mathbf{x}_0$ and agrees with f_S there. If $\mathbf{x}_0 \in \operatorname{Int} S$ or $\operatorname{Ext} S$, f_S and f_A agree near $\mathbf{x}_0$. Let $\mathbf{x}_0 \in \operatorname{Bd} S$. Arbitrarily near $\mathbf{x}_0$ are points $\mathbf{x} \notin S$, where $f_S(\mathbf{x}) = 0$; continuity of f_S at $\mathbf{x}_0$ forces $f_S(\mathbf{x}_0) = 0$. Since $f_A(\mathbf{x})$ is either $f_S(\mathbf{x})$ or 0, $f_A(\mathbf{x}) \to 0 = f_S(\mathbf{x}_0)$; and $f_A(\mathbf{x}_0) = 0$ because $\mathbf{x}_0 \notin A$.

Step 2. If f is integrable over S, f_S is continuous except on a set D of measure zero; by Step 1 so is f_A, so f is integrable over A. Since $f_S - f_A$ vanishes at points not in D, $\int_Q (f_S - f_A) = 0$ by Theorem 11.3, i.e. $\int_S f = \int_A f$. ■

§14. Rectifiable Sets

Motivation. A set has a volume when its characteristic function is integrable, i.e. when its boundary has measure zero. Rectifiable sets are the domains over which integration behaves as expected.

Definition 14.1 (rectifiable set, volume)

Let S be a bounded set in $\mathbb{R}^n$. If the constant function 1 is integrable over S, S is **rectifiable**, and its (n-dimensional) **volume** is

$$\mathrm{v}(S) = \int_S 1.$$

This agrees with the earlier definition when S is a rectangle.

Theorem 14.1

A subset S of $\mathbb{R}^n$ is rectifiable if and only if S is bounded and $\operatorname{Bd} S$ has measure zero.

Proof. The function 1_S (equal to 1 on S and 0 outside) is continuous on the open sets $\operatorname{Ext} S$ and $\operatorname{Int} S$ and fails to be continuous at each point of $\operatorname{Bd} S$. By Theorem 11.2, 1_S is integrable over a rectangle $Q \supset S$ iff $\operatorname{Bd} S$ has measure zero. ■

Theorem 14.2

(a) (Positivity) If S is rectifiable, $\mathrm{v}(S) \geq 0$.

(b) (Monotonicity) If S_1 and S_2 are rectifiable and if $S_1 \subset S_2$, then $\mathrm{v}(S_1) \leq \mathrm{v}(S_2)$.

(c) (Additivity) If S_1 and S_2 are rectifiable, so are $S_1 \cup S_2$ and $S_1 \cap S_2$, and $\mathrm{v}(S_1 \cup S_2) = \mathrm{v}(S_1) + \mathrm{v}(S_2) - \mathrm{v}(S_1 \cap S_2)$.

(d) Suppose S is rectifiable. Then $\mathrm{v}(S) = 0$ if and only if S has measure zero.

(e) If S is rectifiable, so is the set $A = \operatorname{Int} S$, and $\mathrm{v}(S) = \mathrm{v}(A)$.

(f) If S is rectifiable, and if $f : S \to \mathbb{R}$ is a bounded continuous function, then f is integrable over S.

Proof. (a), (b), (c) follow from Theorem 13.3.

(d) follows by applying Theorem 11.3 to the non-negative function 1_S.

(e) follows from Theorem 13.6, and (f) from Theorem 13.5 (here $E \subset \operatorname{Bd} S$, which has measure zero). ■

Terminology

Classically, this volume is called *content* (Jordan content), as opposed to *measure* (Lebesgue measure), the basis of the Lebesgue integral. Measure is defined for more sets, and the two agree when both are defined; a "set of measure zero" as defined here is exactly a set of Lebesgue measure zero, which need not be rectifiable. Munkres uses "rectifiable" for sets having volume (content).

Rectifiable sets are not easy to describe other than by Theorem 14.1. Bounded open or bounded closed sets need not be rectifiable:

Example 14.1

A bounded open set $A \subset \mathbb{R}$ such that $\operatorname{Bd} A$ does not have measure zero. Enumerate the rationals of $(0,1)$ as $q_1, q_2, \ldots$. Fix $0 < a < 1$ and for each i choose an open interval $(a_i, b_i) \subset (0,1)$ of length less than $a/2^i$ containing q_i. Let $A = \bigcup_i (a_i, b_i)$. Suppose $\operatorname{Bd} A$ has measure zero; let $\epsilon = 1 - a$ and cover $\operatorname{Bd} A$ by countably many open intervals of total length less than ϵ. Since $A \subset [0,1]$ contains every rational of $(0,1)$, $\bar{A} = [0,1]$; and $\bar{A} = A \cup \operatorname{Bd} A$. So these intervals, together with the intervals (a_i, b_i), form an open covering of $[0,1]$ of total length less than $\epsilon + \sum a/2^i = \epsilon + a = 1$. By compactness finitely many cover $[0,1]$, contradicting Corollary 10.5.

Definition 14.2 (simple region)

Let C be a compact rectifiable set in $\mathbb{R}^{n-1}$ and $\phi, \psi : C \to \mathbb{R}$ continuous with $\phi(\mathbf{x}) \le \psi(\mathbf{x})$ for $\mathbf{x} \in C$. The set

$$S = \{(\mathbf{x}, t) : \mathbf{x} \in C \text{ and } \phi(\mathbf{x}) \le t \le \psi(\mathbf{x})\}$$

is a **simple region** in $\mathbb{R}^n$. Nothing is special about the last coordinate: if $k + l = n - 1$ and $\mathbf{y}, \mathbf{z}$ denote general points of $\mathbb{R}^k, \mathbb{R}^l$, the set $\{(\mathbf{y}, t, \mathbf{z}) : (\mathbf{y}, \mathbf{z}) \in C,\ \phi(\mathbf{y}, \mathbf{z}) \le t \le \psi(\mathbf{y}, \mathbf{z})\}$ is also called a simple region.

Lemma 14.3

If S is a simple region in $\mathbb{R}^n$, then S is compact and rectifiable.

Proof.

Step 1. Let $G_\phi = \{(\mathbf{x}, t) : \mathbf{x} \in C,\ t = \phi(\mathbf{x})\}$ be the graph of ϕ, similarly G_ψ, and $D = \{(\mathbf{x}, t) : \mathbf{x} \in \operatorname{Bd} C,\ \phi(\mathbf{x}) \le t \le \psi(\mathbf{x})\}$. We show $\operatorname{Bd} S \subset G_\phi \cup G_\psi \cup D$. Each of these lies in S, so then $\operatorname{Bd} S \subset S$, S is closed, and being bounded, compact. Suppose $(\mathbf{x}_0, t_0)$ lies in none of the three sets. There are three cases:

(1) $\mathbf{x}_0 \notin C$;

(2) $\mathbf{x}_0 \in C$ and $t_0 < \phi(\mathbf{x}_0)$ or $t_0 > \psi(\mathbf{x}_0)$;

(3) $\mathbf{x}_0 \in \operatorname{Int} C$ and $\phi(\mathbf{x}_0) < t_0 < \psi(\mathbf{x}_0)$. In case (1) a neighborhood U of $\mathbf{x}_0$ is disjoint from C (C is closed), so $U \times \mathbb{R}$ is disjoint from S and $(\mathbf{x}_0, t_0) \in \operatorname{Ext} S$. In case (2), say $t_0 < \phi(\mathbf{x}_0)$: by continuity of ϕ there is a neighborhood W of $(\mathbf{x}_0, t_0)$ on which $\phi(\mathbf{x}) - t > 0$ for $\mathbf{x} \in C$; W is disjoint from S, so $(\mathbf{x}_0, t_0) \in \operatorname{Ext} S$; similarly if $t_0 > \psi(\mathbf{x}_0)$. In case (3) there is a neighborhood $U \times V$ of $(\mathbf{x}_0, t_0)$ with $U \subset C$ on which $t - \phi(\mathbf{x})$ and $\psi(\mathbf{x}) - t$ are positive; then $U \times V \subset S$ and $(\mathbf{x}_0, t_0) \in \operatorname{Int} S$.

Step 2. G_ϕ (and likewise G_ψ) has measure zero. Choose a rectangle $Q \supset C$ in $\mathbb{R}^{n-1}$. Given ϵ, let $\epsilon' = \epsilon / 2\,\mathrm{v}(Q)$. By uniform continuity of ϕ on the compact set C there is δ with $|\phi(\mathbf{x}) - \phi(\mathbf{y})| < \epsilon'$ for $\mathbf{x}, \mathbf{y} \in C$, $|\mathbf{x} - \mathbf{y}| < \delta$. Choose a partition P of Q of mesh less than δ. For each subrectangle R meeting C pick $\mathbf{x}_R \in R \cap C$ and let $I_R = [\phi(\mathbf{x}_R) - \epsilon', \phi(\mathbf{x}_R) + \epsilon']$. Then $R \times I_R$ contains every point $(\mathbf{x}, \phi(\mathbf{x}))$ with $\mathbf{x} \in C \cap R$, so these rectangles cover G_ϕ, with total volume $\sum_R \mathrm{v}(R)(2\epsilon') \le 2\epsilon'\,\mathrm{v}(Q) = \epsilon$.

Step 3. D has measure zero. Since ϕ, ψ are continuous on the compact set C, there is M with $-M \le \phi(\mathbf{x}) \le \psi(\mathbf{x}) \le M$. Given ϵ, cover $\operatorname{Bd} C$ (measure zero, as C is rectifiable) by rectangles $Q_1, Q_2, \ldots$ in $\mathbb{R}^{n-1}$ of total volume less than $\epsilon/2M$; then the rectangles $Q_i \times [-M, M]$ cover D with total volume less than ϵ. ∎

Theorem 14.4 (Fubini's theorem for simple regions)

Let $S = \{(\mathbf{x}, t) : \mathbf{x} \in C \text{ and } \phi(\mathbf{x}) \le t \le \psi(\mathbf{x})\}$ be a simple region in $\mathbb{R}^n$. Let $f : S \to \mathbb{R}$ be a continuous function. Then f is integrable over S, and

$$\int_S f = \int_{\mathbf{x}\in C} \int_{t=\phi(\mathbf{x})}^{t=\psi(\mathbf{x})} f(\mathbf{x}, t).$$

Proof. Let $Q \times [-M, M]$ be a rectangle in $\mathbb{R}^n$ containing S. Since f is continuous and bounded on the compact rectifiable set S, it is integrable over S (Theorem 14.2(f)). For fixed $\mathbf{x}_0 \in Q$, the function $t \mapsto f_S(\mathbf{x}_0, t)$ is either identically zero (if $\mathbf{x}_0 \notin C$) or continuous at all but two points of $\mathbb{R}$; so it is integrable over $[-M, M]$, and Corollary 12.3 gives

$$\int_Q f_S = \int_{\mathbf{x}\in Q} \int_{t=-M}^{t=M} f_S(\mathbf{x}, t).$$

The inner integral vanishes if $\mathbf{x} \notin C$; and $f_S(\mathbf{x}, t)$ vanishes unless $\phi(\mathbf{x}) \le t \le \psi(\mathbf{x})$, in which case it equals $f(\mathbf{x}, t)$. So $\int_S f = \int_{\mathbf{x}\in C} \int_{t=\phi(\mathbf{x})}^{t=\psi(\mathbf{x})} f(\mathbf{x}, t)$. ∎

If S is not a simple region, one can often write it as a union of simple regions overlapping in sets of measure zero and use Corollary 13.4.

Example 14.2

A set $S \subset \mathbb{R}^2$ that is not a simple region (e.g. a "notched" polygon) can often be broken by a vertical line into simple regions overlapping in a set of measure zero; one then integrates over each piece and adds.

Example 14.3

The annulus $S = \{(x, y) : 1 \le x^2 + y^2 \le 4\}$ is not a simple region, but it is the union of two simple regions (its upper and lower halves) overlapping in a set of measure zero. The limits of integration are unpleasant; in practice one passes to polar coordinates—a special case of the change of variables theorem (Chapter 4).

Volume and rigid motions

Our discussion lacks one thing: why is volume independent of position? If $h : \mathbb{R}^n \to \mathbb{R}^n$ satisfies $\|h(\mathbf{x}) - h(\mathbf{y})\| = \|\mathbf{x} - \mathbf{y}\|$ (an **isometry**) and S is rectifiable, is $T = h(S)$ rectifiable with $\mathrm{v}(T) = \mathrm{v}(S)$? For instance, the square S with vertices $(0,0), (5,0), (5,5), (0,5)$ rotated through $\theta = \arctan\frac{3}{4}$ is the square T with vertices $(0,0), (4,3), (1,7), (-3,4)$. T is a simple region, but showing $\mathrm{v}(T) = 25$ directly (integrating $\psi - \phi$ over $[-3, 4]$, or estimating $L(1_T, P) \le 25 \le U(1_T, P)$) is unappealing. The general result, proved in Chapter 4 (Theorem 20.7), is that isometries preserve volume.

§15. Improper Integrals

Motivation. Many natural integrals live on unbounded sets or have unbounded integrands, for example $\int_{\mathbb{R}^2} e^{-(x^2+y^2)}$. The extended integral over open sets handles them and is the version used in the change of variables theorem.

We now define $\int_S f$ when S or f is unbounded (an *improper integral*), but only for S open in $\mathbb{R}^n$.

Definition 15.1 (extended integral)

Let A be open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ continuous. If $f \geq 0$ on A, the (**extended**) **integral** $\int_A f$ is the supremum of the numbers $\int_D f$ as D ranges over all compact rectifiable subsets of A, provided this supremum exists; then f is **integrable over A (in the extended sense)**. For general continuous f let

$$f_+(\mathbf{x}) = \max\{f(\mathbf{x}), 0\}, \qquad f_-(\mathbf{x}) = \max\{-f(\mathbf{x}), 0\};$$

f is integrable over A if both f_+ and f_- are, and then $\int_A f = \int_A f_+ - \int_A f_-$.

If A and f are bounded, $\int_A f$ could mean the extended or the ordinary integral. If the ordinary integral exists, so does the extended one and they agree (Theorem 15.4), but the extended integral may exist when the ordinary one does not.

Convention

If A is an open set in $\mathbb{R}^n$, then $\int_A f$ will denote the extended integral unless specifically stated otherwise.

Lemma 15.1

Let A be an open set in $\mathbb{R}^n$. Then there exists a sequence $C_1, C_2, \ldots$ of compact rectifiable subsets of A whose union is A, such that $C_N \subset \operatorname{Int} C_{N+1}$ for each N.

Proof. Let d be the sup metric and $B = \mathbb{R}^n - A$. For each positive integer N let

$$D_N = \{\mathbf{x} : d(\mathbf{x}, B) \geq 1/N \text{ and } d(\mathbf{x}, \mathbf{0}) \leq N\}.$$

Since $d(\mathbf{x}, B)$ and $d(\mathbf{x}, \mathbf{0})$ are continuous (proof of Theorem 4.6), D_N is closed; it lies in the cube of radius N, so it is compact. $D_N \subset A$, since $d(\mathbf{x}, B) \geq 1/N$ excludes $\mathbf{x} \in B$. The D_N cover A: for $\mathbf{x} \in A$, $d(\mathbf{x}, B) > 0$ (as A is open), so $d(\mathbf{x}, B) \geq 1/N$ and $d(\mathbf{x}, \mathbf{0}) \leq N$ for large N. The set

$$A_{N+1} = \{\mathbf{x} : d(\mathbf{x}, B) > 1/(N+1) \text{ and } d(\mathbf{x}, \mathbf{0}) < N+1\}$$

is open, contained in D_{N+1}, and contains D_N; hence $D_N \subset \operatorname{Int} D_{N+1}$.

The D_N may not be rectifiable, so we modify them. For each $\mathbf{x} \in D_N$ choose a closed cube centered at $\mathbf{x}$ and contained in $\operatorname{Int} D_{N+1}$. The interiors of these cubes cover D_N; choose finitely many whose interiors cover D_N and let C_N be their union. As a finite union of rectangles, C_N is compact and rectifiable, and $D_N \subset \operatorname{Int} C_N \subset C_N \subset \operatorname{Int} D_{N+1}$. Hence $\bigcup C_N = A$ and $C_N \subset \operatorname{Int} C_{N+1}$. ■

Theorem 15.2

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be continuous. Choose a sequence C_N of compact rectifiable subsets of A whose union is A such that $C_N \subset \operatorname{Int} C_{N+1}$ for each N. Then f is integrable over A if and only if the sequence $\int_{C_N} |f|$ is bounded. In this case,

$$\int_A f = \lim_{N\to\infty} \int_{C_N} f.$$

In particular, f is integrable over A iff $|f|$ is.

Proof.

Step 1. Let $f \geq 0$, so $f = |f|$. The sequence $\int_{C_N} f$ is increasing (monotonicity), so it converges iff it is bounded. If f is integrable over A, then since each C_N is a compact rectifiable subset of A,

$\int_{C_N} f \le \int_A f$; so the sequence is bounded and $\lim \int_{C_N} f \le \int_A f$. Conversely, suppose the sequence is bounded, and let D be any compact rectifiable subset of A. D is covered by the open sets $\operatorname{Int} C_1 \subset \operatorname{Int} C_2 \subset \cdots$, hence by finitely many, hence by one, $\operatorname{Int} C_M$. Then $\int_D f \le \int_{C_M} f \le \lim_N \int_{C_N} f$. So f is integrable over A and $\int_A f \le \lim \int_{C_N} f$.

Step 2. For arbitrary continuous f: f is integrable over A iff f_+ and f_- are, iff the sequences $\int_{C_N} f_+$, $\int_{C_N} f_-$ are bounded (Step 1). Since $0 \le f_\pm \le |f|$ and $|f| = f_+ + f_-$, this holds iff $\int_{C_N} |f|$ is bounded. Then $\int_{C_N} f_\pm \to \int_A f_\pm$, and $\int_{C_N} f = \int_{C_N} f_+ - \int_{C_N} f_-$ converges to $\int_A f_+ - \int_A f_- = \int_A f$. ■

Theorem 15.3

Let A be an open set in $\mathbb{R}^n$. Let $f, g : A \to \mathbb{R}$ be continuous functions.

(a) (Linearity) If f and g are integrable over A, so is $af+bg$; and $\int_A (af+bg) = a\int_A f + b\int_A g$.

(b) (Comparison) Let f and g be integrable over A. If $f(\mathbf{x}) \le g(\mathbf{x})$ for $\mathbf{x} \in A$, then $\int_A f \le \int_A g$. In particular, $|\int_A f| \le \int_A |f|$.

(c) (Monotonicity) Assume B is open and $B \subset A$. If f is non-negative on A and integrable over A, then f is integrable over B and $\int_B f \le \int_A f$.

(d) (Additivity) Suppose A and B are open in $\mathbb{R}^n$ and f is continuous on $A \cup B$. If f is integrable on A and B, then f is integrable on $A \cup B$ and $A \cap B$, and $\int_{A\cup B} f = \int_A f + \int_B f - \int_{A\cap B} f$.

Here all integrals are extended integrals.

Proof. Let C_N be compact rectifiable sets with union A and $C_N \subset \operatorname{Int} C_{N+1}$ (Lemma 15.1).

(a) By the properties of the ordinary integral, $\int_{C_N} |af+bg| \le |a| \int_{C_N} |f| + |b| \int_{C_N} |g|$, which is bounded; so $af + bg$ is integrable (Theorem 15.2). Linearity follows by taking limits in $\int_{C_N} (af + bg) = a\int_{C_N} f + b\int_{C_N} g$.

(b) Take limits in $\int_{C_N} f \le \int_{C_N} g$.

(c) A compact rectifiable subset D of B is one of A, so $\int_D f \le \int_A f$; hence f is integrable over B with $\int_B f \le \int_A f$.

(d) Let D_N be a corresponding sequence for B, and $E_N = C_N \cup D_N$, $F_N = C_N \cap D_N$. These are compact rectifiable with unions $A \cup B$ and $A \cap B$. Also $E_N \subset \operatorname{Int} E_{N+1}$: if $\mathbf{x} \in C_N$, a neighborhood of $\mathbf{x}$ lies in $C_{N+1} \subset E_{N+1}$; similarly for D_N. And $F_N \subset \operatorname{Int} F_{N+1}$: for $\mathbf{x} \in F_N$ there are neighborhoods $U \subset C_{N+1}$, $V \subset D_{N+1}$, so $U \cap V \subset F_{N+1}$. By additivity of the ordinary integral,

$$\int_{E_N} f = \int_{C_N} f + \int_{D_N} f - \int_{F_N} f. \qquad (*)$$

Applying $(*)$ to $|f|$, $\int_{E_N} |f|$ and $\int_{F_N} |f|$ are bounded by $\int_{C_N} |f| + \int_{D_N} |f|$. So f is integrable over $A \cup B$ and $A \cap B$, and the formula follows by taking limits in $(*)$. ■

Theorem 15.4

Let A be a bounded open set in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be a bounded continuous function. Then the extended integral $\int_A f$ exists. If the ordinary integral $\int_A f$ also exists, then these two integrals are equal.

Proof. Let Q be a rectangle containing A.

Step 1. Choose M with $|f| \le M$ on A. For any compact rectifiable $D \subset A$, $\int_D |f| \le \int_D M \le M\,\mathrm{v}(Q)$. So f is integrable over A in the extended sense.

Step 2. Let $f \geq 0$ and suppose the ordinary integral exists; it equals $\int_Q f_A$. For compact rectifiable $D \subset A$, $\int_D f = \int_D f_A \leq \int_Q f_A$ by monotonicity, so (extended) $\int_A f \leq$ (ordinary) $\int_A f$. On the other hand, let P be a partition of Q, let $R_1, \ldots, R_k$ be the subrectangles determined by P that lie in A, and $D = R_1 \cup \cdots \cup R_k$. Since $m_R(f_A) = m_R(f)$ for $R \subset A$ and $m_R(f_A) \leq 0$ otherwise (as f_A vanishes at some point of R),

$$L(f_A, P) \leq \sum_{i=1}^{k} m_{R_i}(f)\, \mathrm{v}(R_i) \leq \sum_{i=1}^{k} \int_{R_i} f = \int_D f \leq (\text{extended}) \int_A f,$$

using comparison and additivity (Corollary 13.4). As P is arbitrary, (ordinary) $\int_A f \leq$ (extended) $\int_A f$.

Step 3. In general, if the ordinary integral exists, so do the ordinary integrals of f_+, f_- (Lemma 13.2), and (ordinary) $\int_A f =$ (ordinary) $\int_A f_+ -$ (ordinary) $\int_A f_- =$ (extended) $\int_A f_+ -$ (extended) $\int_A f_- =$ (extended) $\int_A f$. ■

Example 15.1

If A is a bounded open set and f a bounded continuous function on A, the extended integral exists but the ordinary one may not. With A the open subset of $\mathbb{R}$ of Example 14.1, Bd A does not have measure zero, so the ordinary integral $\int_A 1$ does not exist (Theorem 14.1), although the extended integral $\int_A 1$ does.

Corollary 15.5

Let S be a bounded set in $\mathbb{R}^n$; let $f : S \to \mathbb{R}$ be a bounded continuous function. If f is integrable over S in the ordinary sense, then

$$(\text{ordinary}) \int_S f = (\text{extended}) \int_{\operatorname{Int} S} f.$$

Proof. Apply Theorems 13.6 and 15.4. ■

So theorems about extended integrals have consequences for ordinary integrals; the change of variables theorem (Chapter 4) is an important example. A third formulation of the extended integral is given in Chapter 4 (via partitions of unity). The following one is often convenient in practice:

Theorem 15.6

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be continuous. Let $U_1 \subset U_2 \subset \cdots$ be a sequence of open sets whose union is A. Then $\int_A f$ exists if and only if the sequence $\int_{U_N} |f|$ exists and is bounded; in this case,

$$\int_A f = \lim_{N \to \infty} \int_{U_N} f.$$

Proof. As usual it suffices to consider $f \geq 0$. If $\int_A f$ exists, monotonicity (Theorem 15.3(c)) shows f is integrable over each U_N with $\int_{U_N} f \leq \int_A f$; so the increasing sequence $\int_{U_N} f$ converges, and $\lim \int_{U_N} f \leq \int_A f$. Conversely, suppose $\int_{U_N} f$ exists and is bounded. A compact rectifiable $D \subset A$ is covered by $U_1 \subset U_2 \subset \cdots$, hence by one U_M; by definition $\int_D f \leq \int_{U_M} f \leq \lim_N \int_{U_N} f$. As D is arbitrary, $\int_A f \leq \lim_N \int_{U_N} f$. ■

Usually one chooses U_N rectifiable with f bounded on U_N; then $\int_{U_N} f$ exists as an ordinary integral (hence as an extended one) and can be computed by familiar techniques.

Example 15.2

Let $A = \{(x, y) : x > 1,\ y > 1\}$ and $f(x, y) = 1/x^2y^2$; f is bounded on A but A is unbounded. Let $U_N = (1, N)^2$; U_N is rectifiable, f is bounded and continuous on $\bar{U}_N$, so $\int_{U_N} f$ exists as an ordinary integral and Fubini's theorem applies:

$$\int_{U_N} f = \int_{x=1}^{x=N} \int_{y=1}^{y=N} 1/x^2y^2 = ((N-1)/N)^2.$$

Hence $\int_A f = 1$. (One could also use Theorem 15.2 with $C_N = [(N+1)/N, N]^2$.)

Example 15.3

Let $B = (0, 1)^2$ and $f(x, y) = 1/x^2y^2$. Now B is bounded but f is unbounded near each point of the x- and y-axes. With $U_N = (1/N, 1)^2$, f is bounded on U_N and

$$\int_{U_N} f = (-1 + N)^2,$$

which is unbounded. Hence $\int_B f$ does not exist.